Keys Types : set up relations amongst various columns and tables of a relational database.

1. **Super key** : The set of one or more attributes (columns) that can uniquely identified a tuple (record) is known as Super key.

A super key is a candidate key's superset.

Example : (Employee_ID), (Email) and (Employee_ID, Email) are all Super key.

2. **Candidate key** : A set of columns that can uniquely identify each row; one is chosen as the primary key.

Example:

```
CREATE TABLE Employee {
    Employee_ID   INT,
    Email   VARCHAR(100),
    Phone_Num   VARCHAR(15),
    Primary key (Employee_ID)
);
```

Here, Employee_ID and Email Candidate key.

3. Primary key : A column or a set of columns that uniquely identifies each row in a table.

Example :

```
CREATE TABLE Students (
    Student_ID Int Primary key,
    Name VARCHAR (50));
```

Here, Student_ID Primary key.

4. Foreign key : A column that refers to the primary key of another table to establish relationships between tables.

Example :

```
CREATE TABLE Courses (
    Course_ID IND Primary key,
    Course_Name VARCHAR (100),
    Student_ID INT,
    FOREIGN key (student_ID)
    References Students (Student_ID)
);
```

Here, Student_ID in the Course Table is a foreign key.

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5. <u>Alternate key</u>: Candidate keys that aren't selected as the primary key.

<u>Example</u>: If Employee_ID is the Primary key, then Email is an Alternate key.

6. <u>Composite key</u>: A key made up of two or more columns to uniquely identify a record.

<u>Example</u>:

```
CRATE TABLE Orders (
    Order_ID INT,
    Product_ID INT,
    Primary key (Order_ID, Product_ID)
);
```

Here, Order_ID and Product_ID together form a Composite Primary key.

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